EXPERIMENT 1 – GIT BASH

Repository initialization, GitHub push, branching, multiplication table, and greatest of two numbers

Aim

To initialize a Git repository for a data science project, publish it on GitHub, create and commit Python programs, create the GIT-EXAM branch, and compare the main and GIT-EXAM branches.

Requirements

• Windows computer with Git Bash and VS Code installed.

• GitHub account and internet connection.

• Python 3 installed and available in the terminal.

Where to type each item

Location	Use it for
Git Bash	All Git commands: configuration, init, remote, add, commit, push, branch, diff, log, switch.
VS Code	Create and save program1.py and program2; run the Python programs in the VS Code terminal.
Web browser	Create the empty GitHub repository and verify the pushed files/branches.

PROCEDURE – STEP BY STEP

Step 1 – Check Git installation

Copy/paste in: Git Bash

```
git --version
```

Step 2 – Configure your Git username

Copy/paste in: Git Bash

```
git config --global user.name "Omkar-Munde"
git config --global user.name
```

Step 3 – Configure your Git email

Copy/paste in: Git Bash

```
git config --global user.email "omkarmunde2121@gmail.com"
git config --global user.email
```

Step 4 – Set the default branch name to main

Copy/paste in: Git Bash

```
git config --global init.defaultBranch main
git config --global --list
```

Step 5 – Create the project folder

Copy/paste in: Git Bash

```
cd ~/Documents
mkdir DataScienceProject
cd DataScienceProject
```

If you want the folder in another location, create it there instead. In this practical, the project folder used was C:\Users\Omkar\DataScienceProject.

Step 6 – Initialize the Git repository

Copy/paste in: Git Bash

```
git init
git status
```

Expected status: On branch main; No commits yet; nothing to commit. If Git initializes a different branch, run: git branch -M main

Step 7 – Create an empty repository on GitHub

Copy/paste in: Web browser

Open GitHub and sign in. Select + → New repository. Enter DataScienceProject as the repository name. Choose Public or Private as instructed. Leave Add a README, .gitignore, and license unchecked. Click Create repository.

Step 8 – Connect the local repository to GitHub

Copy/paste in: Git Bash

```
git remote add origin https://github.com/Omkar-Munde/DataScienceProject.git
git remote -v
```

Expected remote output should show the repository URL for both fetch and push. If origin already exists, use: git remote set-url origin https://github.com/Omkar-Munde/DataScienceProject.git

Step 9 – Open the project folder in VS Code

Copy/paste in: Git Bash

```
code .
```

If code . does not work, open VS Code → File → Open Folder → select DataScienceProject.

Step 10 – Create program1.py

Copy/paste in: VS Code

In the Explorer panel, select New File and name it program1.py. Paste the following code and save with Ctrl+S:

```
number = int(input("Enter a number: "))

print("Multiplication table of", number)

for i in range(1, 11):
    print(number, "x", i, "=", number * i)
```

Step 11 – Run program1.py

Copy/paste in: VS Code Terminal or Git Bash

```
python program1.py
```

Example: enter 5. The output should show the multiplication table from 5 x 1 = 5 through 5 x 10 = 50.

Step 12 – Check the status before staging

Copy/paste in: Git Bash

```
git status
```

Step 13 – Stage program1.py

Copy/paste in: Git Bash

```
git add program1.py
git status
```

Expected status should show program1.py under Changes to be committed.

Step 14 – Commit program1.py

Copy/paste in: Git Bash

```
git commit -m "Add multiplication table program"
```

Step 15 – Push the main branch to GitHub

Copy/paste in: Git Bash

```
git push -u origin main
```

Complete GitHub browser authentication if prompted. Then open the GitHub repository and confirm program1.py is visible on main.

Step 16 – Create and switch to GIT-EXAM

Copy/paste in: Git Bash

```
git switch -c GIT-EXAM
git branch
```

The asterisk (*) should appear next to GIT-EXAM, showing that it is the current branch.

Step 17 – Create the second file named program2

Copy/paste in: VS Code

In the VS Code Explorer, select New File and name it exactly program2 (without .py, as specified in the practical). Paste the code below and save with Ctrl+S:

```
num1 = float(input("Enter the first number: "))
num2 = float(input("Enter the second number: "))

if num1 > num2:
    print("Greatest number is", num1)
elif num2 > num1:
    print("Greatest number is", num2)
else:
    print("Both numbers are equal")
```

Step 18 – Run program2

Copy/paste in: VS Code Terminal or Git Bash

```
python program2
```

Example: enter 21 and 54. The program should display: Greatest number is 54.0. If you named the file program2.py instead, run python program2.py and use that exact filename in the Git commands.

Step 19 – Check status and stage program2

Copy/paste in: Git Bash

```
git status
git add program2
git status
```

Expected status should show program2 under Changes to be committed.

Step 20 – Commit program2

Copy/paste in: Git Bash

```
git commit -m "Add program to find greatest of two numbers"
```

Step 21 – Push GIT-EXAM to GitHub

Copy/paste in: Git Bash

```
git push -u origin GIT-EXAM
```

Complete browser authentication if requested. On GitHub, use the branch dropdown to select GIT-EXAM and confirm that program1.py and program2 are visible.

Step 22 – List local and remote branches

Copy/paste in: Git Bash

```
git branch -a
```

Expected branch list includes main, GIT-EXAM, remotes/origin/main, and remotes/origin/GIT-EXAM.

Step 23 – Display commit history

Copy/paste in: Git Bash

```
git log --oneline --graph --decorate --all
```

The history should show the multiplication-table commit on main and the greatest-of-two-numbers commit on GIT-EXAM.

Step 24 – Compare the branches

Copy/paste in: Git Bash

```
git diff --name-status main..GIT-EXAM
```

Expected output: A program2. A means Added; program2 is present in GIT-EXAM but not in main.

Step 25 – Verify files on main

Copy/paste in: Git Bash

```
git switch main
ls
```

Expected output: program1.py

Step 26 – Verify files on GIT-EXAM

Copy/paste in: Git Bash

```
git switch GIT-EXAM
ls
```

Expected output: program1.py and program2.

EXPECTED RESULT

The local Git repository is initialized and connected to GitHub. The multiplication-table program is committed and pushed to main. The GIT-EXAM branch is created, the greatest-of-two-numbers program is committed and pushed, and the branch comparison confirms that program2 is an addition on GIT-EXAM.

FINAL FILE STRUCTURE

main branch

```
DataScienceProject/
■■■ program1.py
```

GIT-EXAM branch

```
DataScienceProject/
■■■ program1.py
■■■ program2
```

COMMON ERRORS AND FIXES

remote origin already exists: Run: git remote set-url origin https://github.com/Omkar-Munde/DataScienceProject.git

src refspec main does not match any: Make sure a commit exists and check the branch with: git branch --show-current

Python command not found: Install Python with PATH enabled, then close and reopen Git Bash/VS Code.

Authentication required: Complete GitHub sign-in through the browser/Git Credential Manager. Do not put a password or token in your project files.

GitHub repository: https://github.com/Omkar-Munde/DataScienceProject